

iGen News — Tech Stack & Build Specification

India Global News (IGN) · indiaglobalnews.com · iGenWorld

■ AI-CODING TOOL READY — Structured for Claude Code · Cursor · Bolt · Lovable · Google Jules

■ INSTRUCTIONS FOR AI CODING TOOLS

You are building **India Global News (IGN)** — a B2B news platform at **indiaglobalnews.com**, part of the **iGenWorld** ecosystem.

This document is the **single source of truth** for the entire tech stack, architecture, UI system, and build sequence.

- ◆ Read ALL sections before writing any code.
- ◆ Use the exact technology versions specified. Do not substitute alternatives without noted reason.
- ◆ Follow the Brand Identity section strictly — colours, fonts, and spacing are non-negotiable.
- ◆ Build in the order specified in the Build Sequence section.
- ◆ Each page listed in the Page Registry must be implemented as specified in Step 2 document.
- ◆ All environment variables must use the naming conventions defined in the ENV section.

1. BRAND IDENTITY & DESIGN SYSTEM

1.1 Colour Palette

Indigo Navy	Steel Blue	Saffron Gold	Export Green	White	Light Grey	Dark Grey
#1E3A5F	#2F6FA3	#F4A024	#2E7D32	#FFFFFF	#F5F5F5	#5A6A7A
40%	25%	20%	15%	—	—	—

--color-primary	#1E3A5F	--color-primary	Nav bars, page headers, card headers, primary buttons, footer background (40% of UI)
--color-secondary	#2F6FA3	--color-secondary	Active states, links, secondary buttons, hover states, sub-headers (25% of UI)
--color-accent-gold	#F4A024	--color-accent-gold	CTAs (Upgrade Now, Subscribe), trending badges, breaking news ticker, highlights ONLY (20%)
--color-accent-green	#2E7D32	--color-accent-green	Success states, Export/IGE cross-nav elements, verified badges, save confirmations (15%)
--color-bg-white	#FFFFFF	--color-bg-white	Primary page background, card backgrounds, modal backgrounds
--color-bg-light	#F5F5F5	--color-bg-light	Alternate row backgrounds, section dividers, sidebar backgrounds
--color-text-dark	#5A6A7A	--color-text-dark	Secondary text, metadata, timestamps, placeholder text
--color-text-body	#1A1A1A	--color-text-body	Primary body text, article content, headlines

1.2 Typography System

Role	Font Family	Weight	Size (desktop)	Usage
Display / H1	Playfair Display	Bold 700	36–48px	Article main headlines, hero headings — premium news feel
H2 Sub-heading	Playfair Display	SemiBold 600	24–30px	Section titles, article sub-headings

H3 Card Title	Inter	SemiBold 600	16–20px	News card headlines, sidebar headings
Body Text	Inter	Regular 400	15–16px	Article body, descriptions, UI text
UI Labels / Nav	Inter	Medium 500	13–14px	Navigation, buttons, form labels, tags
Metadata / Micro	Inter	Regular 400	11–12px	Timestamps, read time, comment counts, footnotes
Code / Monospace	JetBrains Mono	Regular 400	13px	Technical content only (for developer-facing pages)

Note: Load fonts via: Google Fonts — Inter (300,400,500,600,700) + Playfair Display (600,700). Add font-display: swap for performance.

1.3 Spacing, Border Radius & Shadows

--spacing-xs	4px	Icon gaps, chip padding
--spacing-sm	8px	Card internal padding, button padding vertical
--spacing-md	16px	Section internal padding, form field gaps
--spacing-lg	24px	Card-to-card gaps, section margins
--spacing-xl	40px	Page section top/bottom margins
--spacing-2xl	64px	Hero section padding, page top padding
--radius-sm	4px	Chip/tag border radius
--radius-md	8px	Card border radius, button radius
--radius-lg	12px	Modal, sidebar panel, large card radius
--radius-full	999px	Pill buttons, avatar circles, toggle switches
--shadow-card	0 2px 8px rgba(30,58,95,0.08)	Default card shadow
--shadow-hover	0 8px 24px rgba(30,58,95,0.16)	Card hover state shadow
--shadow-modal	0 24px 64px rgba(0,0,0,0.24)	Modal/drawer shadow

2. FULL TECHNOLOGY STACK

2.1 Frontend

FRAMEWORK	Next.js 14 (App Router)	14.x	SSR + SSG + ISR for news pages. App router for layout nesting. SEO-critical.
LANGUAGE	TypeScript	5.x	Strict typing across entire codebase. All components, API calls, and data models typed.
STYLING	Tailwind CSS	3.x	Utility-first. Custom theme configured with brand tokens. No separate CSS files.
UI LIBRARY	shadcn/ui	latest	Headless accessible components: Dialog, Sheet, Tabs, Dropdown, Toast, Badge, Avatar.
ANIMATIONS	Framer Motion	11.x	Page transitions, card hover effects, modal open/close, skeleton loaders.
ICONS	Lucide React	latest	Single icon library. Icons: Bell, Bookmark, Heart, Share2, Search, Menu, ChevronDown.
FORMS	React Hook Form + Zod	latest	Sign-up, login, comment, newsletter forms. Zod schema validation.
STATE	Zustand	4.x	Global state: auth session, reader preferences, bookmarks, notification count.
DATA FETCH	TanStack Query (React Query)	5.x	Server state, caching, infinite scroll for feeds, optimistic updates for likes/bookmarks.
RICH TEXT	TipTap	latest	Article body rendering (CMS output). Comments rich text editor.

2.2 Backend

RUNTIME	Node.js	20 LTS	JavaScript runtime for API server.
FRAMEWORK	Next.js API Routes + tRPC	14.x	Type-safe API routes. tRPC for internal frontend↔backend calls without REST boilerplate.
AUTH	NextAuth.js (Auth.js)	5.x	Google OAuth + LinkedIn OAuth + Email/Password. JWT sessions. Role-based (Free/Reader/Leader/Corporate).
DATABASE ORM	Prisma	5.x	Type-safe DB queries. Schema defines: User, Article, Sector, Country, Leader, Bookmark, Comment, Like, Subscription.
DATABASE	PostgreSQL	16.x	Primary relational DB. Hosted on Supabase or Railway. Stores all structured data.
CACHE	Redis (Upstash)	latest	Feed caching, trending query caching (TTL 5 min), rate limiting, session store.
SEARCH	Meilisearch	latest	Full-text article search. Instant autocomplete. Filters by sector/country/leader/date.
EMAIL	Resend	latest	Transactional emails: verification, password reset, welcome. Newsletter via React Email templates.
FILE/MEDIA	Cloudinary	latest	Article image upload, transformation, optimisation. Auto WebP, responsive sizes.
PAYMENTS	Razorpay	latest	Indian payment gateway. Subscription plans: Free/Reader/Leader/Corporate. UPI + cards.
RSS PIPELINE	Bullmq + Redis	latest	Job queue for RSS AI content pipeline. Fetch → AI rewrite → category tag → DB store.
AI CONTENT	Anthropic Claude API	Sonnet 4.5	RSS AI rewriting, article tagging, content summarisation, "Ask IGN AI" Q&A; feature.

2.3 Infrastructure & DevOps

HOSTING	Vercel	latest	Next.js deployment. Edge functions for ISR. Auto CDN. Preview deployments per PR.
DB HOSTING	Supabase	latest	PostgreSQL managed hosting + realtime subscriptions + built-in auth fallback.
CDN	Cloudflare	latest	DNS, DDoS protection, edge caching, image optimisation proxy.
MONITORING	Sentry	latest	Error tracking frontend + backend. Performance monitoring. Alert on errors.
ANALYTICS	PostHog	latest	Product analytics: page views, funnel analysis, feature flags, session replay.
LOGGING	Axiom	latest	Structured log management. API logs, RSS pipeline logs, payment events.
CI/CD	GitHub Actions	latest	Lint → Test → Build → Deploy pipeline. Branch protection on main.
REPO	GitHub	latest	Monorepo: /apps/web (Next.js) /packages/db (Prisma) /packages/ui (shadcn) /services/rss.

2.4 CMS & Content Tools

CMS	Payload CMS	3.x	Self-hosted headless CMS. Article CRUD, editorial workflow (draft→review→publish), media manager.
CMS HOST	Railway	latest	Payload CMS + PostgreSQL admin instance hosted on Railway (separate from main app).
SEO	next-seo + next-sitemap	latest	Meta tags, OG images, Twitter cards, JSON-LD NewsArticle schema, auto sitemap.
PWA	next-pwa	latest	Service worker, offline caching of last 20 articles, home screen install, push notifications.

3. PROJECT FOLDER STRUCTURE

The project is a monorepo. AI tools must create this exact structure:

```
ter, Sidebar                                news/ # NewsCard, ArticleBody, TagChip
```

4. DATABASE SCHEMA (Prisma)

Core Prisma models. AI tools must implement these in /packages/db/schema.prisma:

```
type ContentType // RSS_AI | MANUAL_AI | SPONSORED | PR | PODCAST                                status
```

5. ENVIRONMENT VARIABLES (.env)

Create .env.local in /apps/web/ with these exact variable names:

```
RAZORPAY_KEY_SECRET="..."
```

6. TAILWIND CSS THEME CONFIGURATION

Add this to tailwind.config.ts — maps brand tokens to Tailwind utility classes:

```
body: ['Inter', 'system-ui', 'sans-serif'], // Body / UI                                mono:
```

7. KEY COMPONENT SPECIFICATIONS

7.1 NewsCard Component Props

```
commentCount: number
```

7.2 MegaMenu Component Structure

```
{ label: 'Explore IGE →', href: process.env.NEXT_PUBLIC_IGE_URL, external: true },                                ]
```

7.3 Feed Page Data Fetching

```
page: 1,                                limit:
```

8. BUILD SEQUENCE — STEP BY STEP FOR AI TOOLS

Follow this exact order. Complete each phase before starting the next.

PHASE 1

Foundation Setup

- ◆ Init monorepo with Turborepo. Create /apps/web (Next.js 14), /packages/db (Prisma), /services/rss.
- ◆ Configure Tailwind with brand tokens (Section 6). Install shadcn/ui. Set up Inter + Playfair Display fonts.
- ◆ Set up PostgreSQL on Supabase. Apply Prisma schema (Section 4). Run first migration.
- ◆ Configure .env.local with all variables (Section 5). Set up GitHub repo + Actions CI.
- ◆ Build persistent layout: Root layout with MegaMenu + Footer. Header bar with all elements.

PHASE 2

Auth & Onboarding

- ◆ Implement NextAuth.js: Google OAuth + LinkedIn OAuth + Email/Password.
- ◆ Build Sign-Up 6-step stepper (/signup): Basic Info → Sector → Industry → Country → Leader → Plan.
- ◆ Build Login page (/login) with OAuth buttons.
- ◆ Implement role-based middleware: Free / Reader / Leader / Corporate plan gating.
- ◆ Build Reader Profile page (/profile) with interest editing.

PHASE 3

Core Pages — Feed, Headlines, Trending

- ◆ Build shared FeedLayout (3-column: LeftPanel + CenterFeed + RightSidebar). Used by all 3 pages.
- ◆ Build Feed page (/feed): SpotlightCard + UpcomingHighlights + NewsCard list + SubTabBar.
- ◆ Build Headlines page (/headlines): Copy Feed layout, swap content logic (likes-weighted ranking).
- ◆ Build Trending page (/trending): Copy Feed layout + add Category Filter Pill Bar.
- ◆ Build NewsCard component with ALL elements (Section 7.1) including Save + Read Later buttons.
- ◆ Implement MegaMenu with all 5 items, sub-menus, and IGE cross-navigation links.

PHASE 4

Categories & Article Pages

- ◆ Build Categories pages (P4/P5/P6): Sector grid, Country pairs grid, Leader designations grid.
- ◆ Implement paywall lock overlays on locked category cards.
- ◆ Build Article Detail page (/article/[slug]): Full article body, engagement bar, comments, related articles.
- ◆ Add Save/Bookmark, Read Later, Like, Comment, Share to article page.
- ◆ Implement Premium Paywall Blur component for free-plan article limitation.
- ◆ Add "Explore IGE →" cross-navigation CTA at end of all articles.

My News, Search & Subscription

- ◆ Build My News page (/mynews) with all 5 sub-tabs: Favourites, Bookmarks, Read Later, Comments, Likes.
- ◆ Build Search page (/search) with Meilisearch integration. Autocomplete. Filters.
- ◆ Build Subscription/Pricing page (/plans) with 4 plan cards + feature comparison table.
- ◆ Integrate Razorpay: Subscription checkout flow, webhook for plan upgrades.
- ◆ Build Upgrade Now modal/prompt (triggered by locked content clicks across all pages).

PHASE 6

RSS AI Pipeline & CMS

- ◆ Deploy Payload CMS on Railway. Configure Article collection with all fields.
- ◆ Build RSS AI Pipeline (/services/rss): BullMQ job queue → fetch RSS → Claude API rewrite → auto-tag → save to DB.
- ◆ Set up cron schedule for RSS pipeline (every 30 minutes).
- ◆ Integrate Meilisearch: Index articles on publish. Sync on update/delete.
- ◆ Build notification system: Web Push via OneSignal. Breaking news triggers. Daily briefing.

PHASE 7

SEO, Performance & Launch Prep

- ◆ Add next-seo to all pages: meta titles, descriptions, OG images, Twitter cards.
- ◆ Add JSON-LD NewsArticle schema to all article pages.
- ◆ Configure next-sitemap: auto-generate sitemap.xml. Submit to Google Search Console.
- ◆ Set up next-pwa: service worker, offline caching, home screen install.
- ◆ Run Lighthouse audit. Target: Performance >90, Accessibility >95, SEO >95.
- ◆ Set up Sentry error tracking. PostHog analytics. Vercel deployment to production.

9. PAGE REGISTRY — ALL ROUTES

/feed	Home / Feed	SSR + ISR	Required	60s
/headlines	Headlines	SSR + ISR	Required	120s
/trending	Trending	SSR + ISR	Public*	60s
/categories/sector	Sector Categories	SSG + ISR	Public	3600s
/categories/country	Country Categories	SSG + ISR	Public	3600s
/categories/leader	Leader Categories	SSG + ISR	Public	3600s
/article/[slug]	Article Detail	SSG + ISR	Public*	300s
/search	Search Results	CSR	Public	live
/mynews	My News	CSR	Required	live
/plans	Subscription Pricing	SSG	Public	daily
/profile	Reader Profile	CSR	Required	live
/signup	Sign Up Flow	CSR	Redirect*	—
/login	Login	CSR	Redirect*	—
/podcasts	Podcast Listing	SSG + ISR	Public	1800s

Note: * Public with paywall: Trending and Article Detail visible to all, but reading full locked content requires auth + paid plan. /signup and /login redirect to /feed if already logged in.

10. API ROUTES REFERENCE

GET	/api/feed	Fetch personalised feed (auth + plan-gated)
GET	/api/headlines	Fetch likes-weighted headlines
GET	/api/trending	Fetch trending (all, sorted by score)
GET	/api/articles/:slug	Single article data
GET	/api/search?q=&tab;=&sort;=	Meilisearch proxy
POST	/api/articles/:id/like	Toggle like (auth required)
POST	/api/articles/:id/bookmark	Toggle bookmark (auth required)
POST	/api/articles/:id/readlater	Add/remove from Read Later
POST	/api/comments	Post new comment

GET	/api/categories/sectors	All 20 sectors with article counts
GET	/api/categories/countries	All 195 country pairs
GET	/api/categories/leaders	All 25 leader designations
GET	/api/user/profile	Reader profile + selections
PUT	/api/user/interests	Update sector/country/leader selections
POST	/api/subscriptions/checkout	Razorpay checkout session
POST	/api/subscriptions/webhook	Razorpay webhook — plan upgrade
POST	/api/newsletter/subscribe	Newsletter subscription via Resend
POST	/api/rss/trigger	Manually trigger RSS AI pipeline run (admin)